Project4

March 25, 2026

[1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
import warnings
warnings.filterwarnings('ignore')
```

1 Load the Data Set

[2]:
```
customer_df = pd.read_csv("C:/Users/emmar/OneDrive/Desktop/Projects/Project 4/
↪Mall_Customers.csv")

# show the first 5 rows
customer_df.head(10)
```

[2]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
5	6	Female	22	17	76
6	7	Female	35	18	6
7	8	Female	23	18	94
8	9	Male	64	19	3
9	10	Female	30	19	72

2 Univariate Analysis

I want to analyze and visualize one variable at a time.

[3]:
```
customer_df.describe()
```

[3]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522

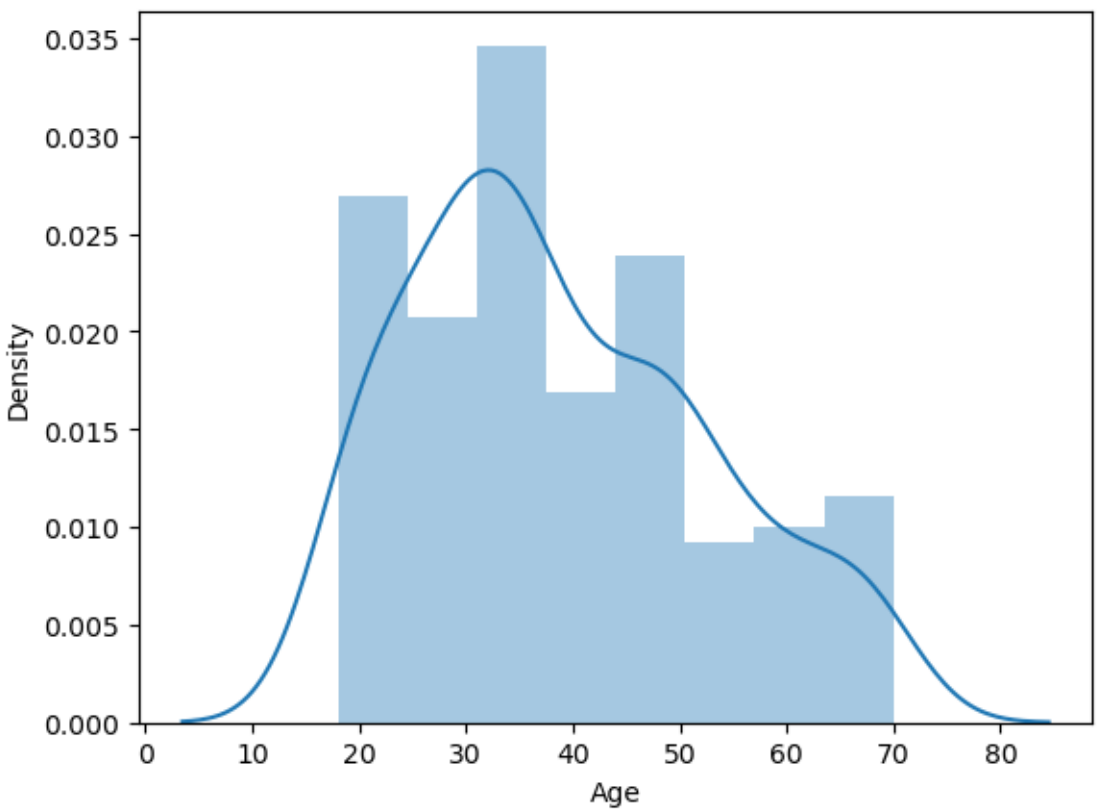
```
min      1.000000   18.000000          15.000000                1.000000
25%     50.750000   28.750000          41.500000               34.750000
50%    100.500000   36.000000          61.500000               50.000000
75%    150.250000   49.000000          78.000000               73.000000
max    200.000000   70.000000         137.000000               99.000000
```

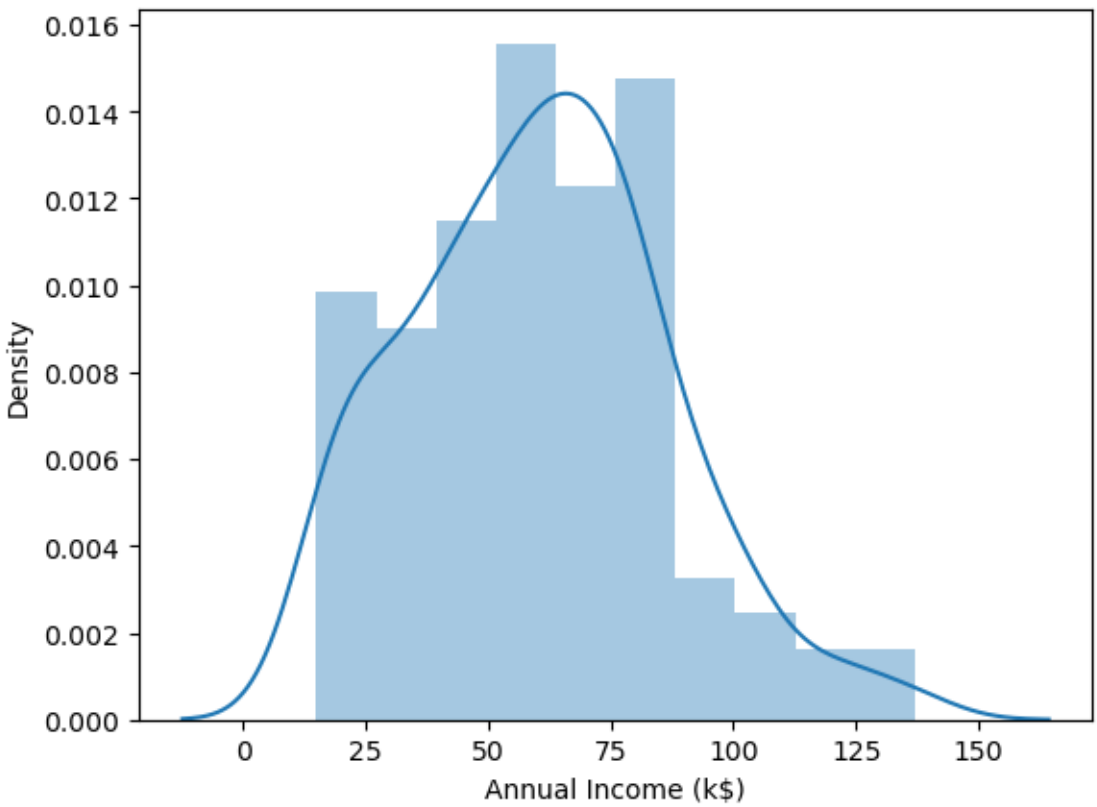
I can see the count, mean, standard deviation, and quartiles for each variable using describe. This helps me understand the dataset more and one thing I noticed is that in every variable the mean and median are very close. This tells me that each variable should have a normal distribution and I will visualize the distributions next to confirm.

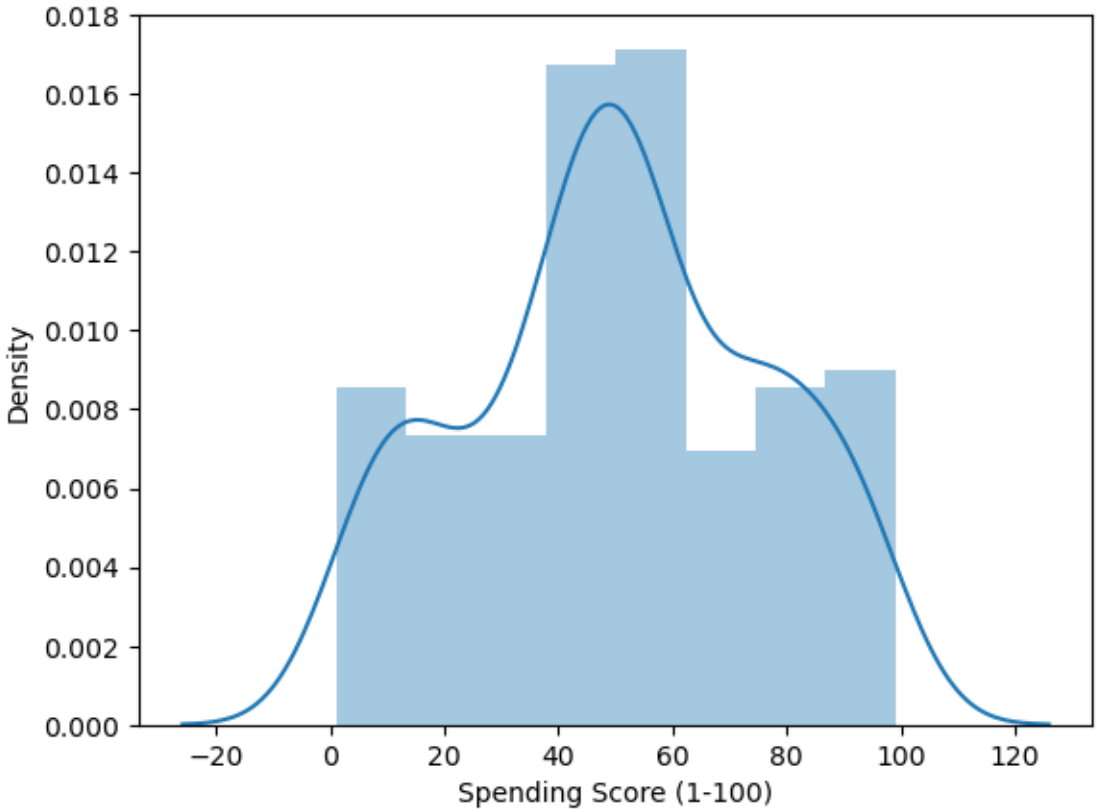
[4]:
```
customer_df.columns
```

[4]:
```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)'],
      dtype='object')
```

[5]:
```
columns = ['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
for i in columns:
    plt.figure()
    sns.distplot(customer_df[i])
```

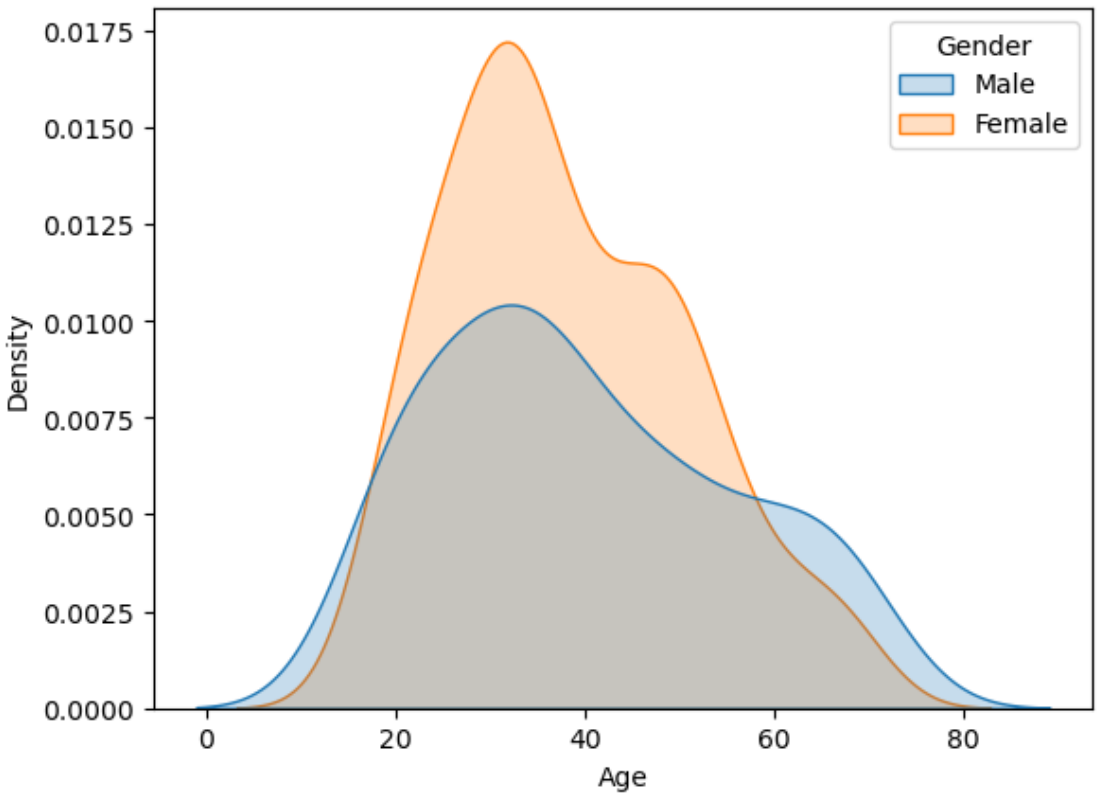


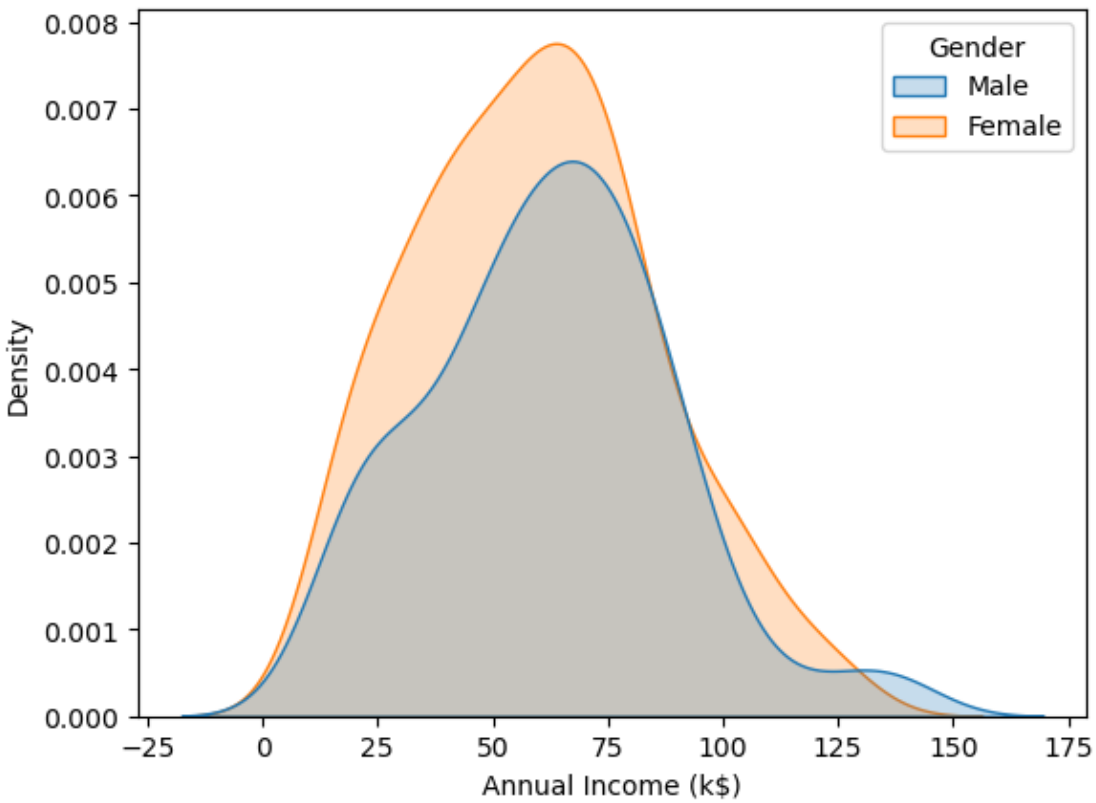

0.016
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
Density
0
25
50
75
100
125
150
Annual Income (k$)

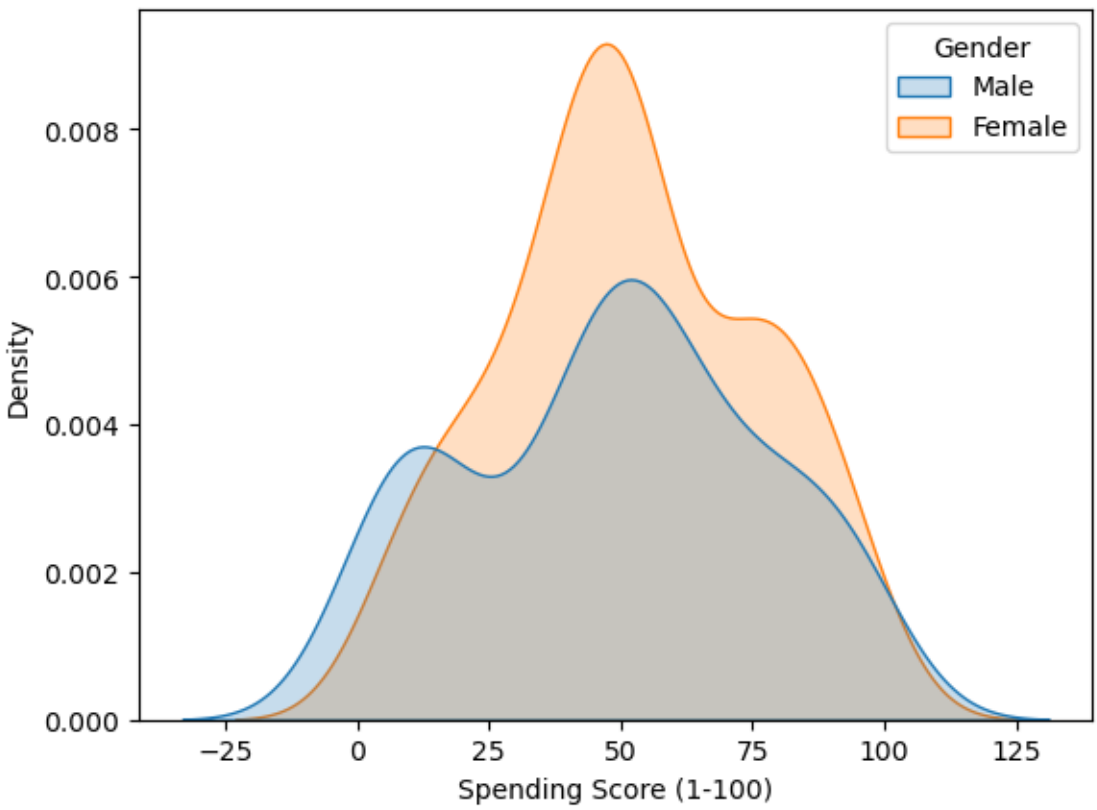


Looking at the visuals I can confirm that everything looks to be normally distributed. The one thing I would like to note is that the spending score is focused bweteen 40 and 60. Now I want to get more information and try to divide the data by gender since gender is in our data set.

[6]:
```
for i in columns:
    plt.figure()
    sns.kdeplot(data=customer_df, x=i, hue='Gender', fill=True)
```

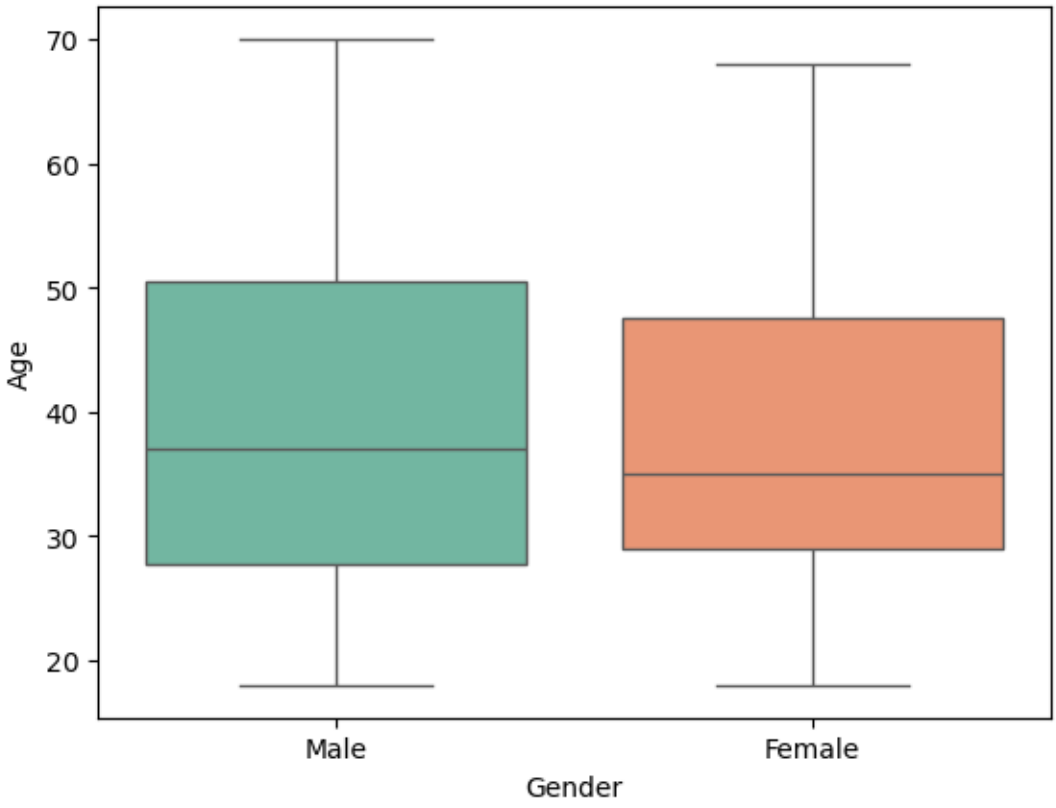

Gender
Male
Female
Density
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0
20
40
60
80
Age

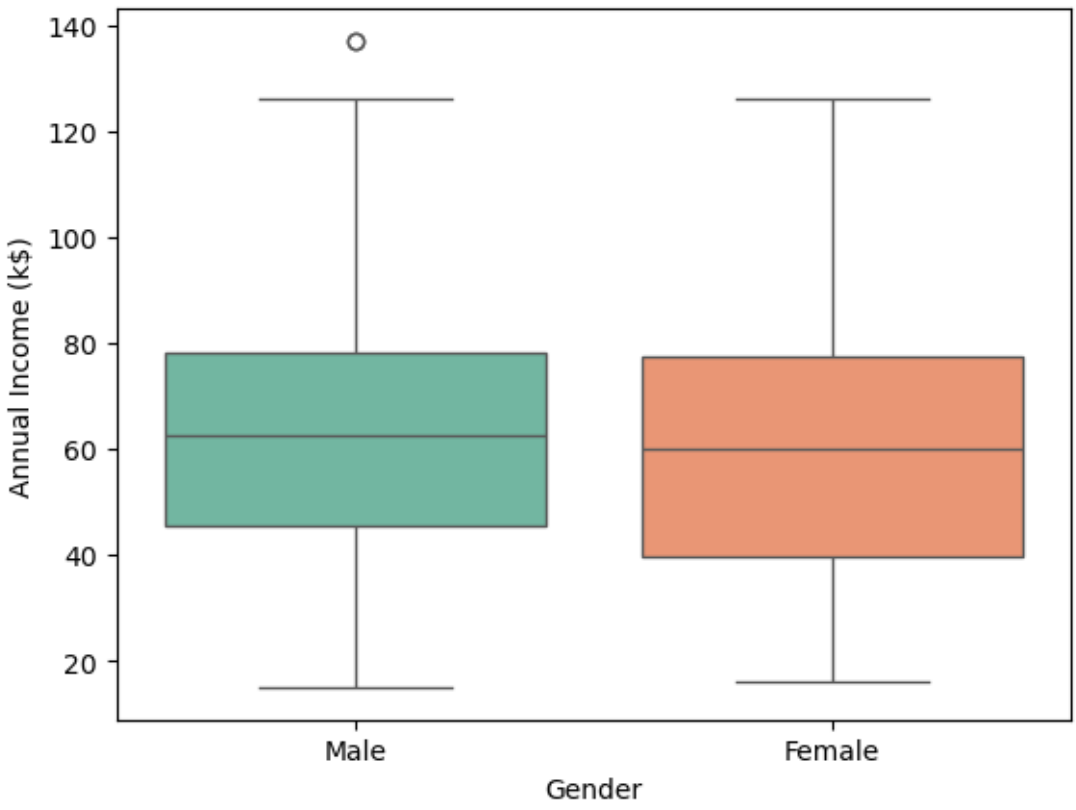

Gender
Male
Female
Density
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−25
0
25
50
75
100
125
150
175
Annual Income (k$)

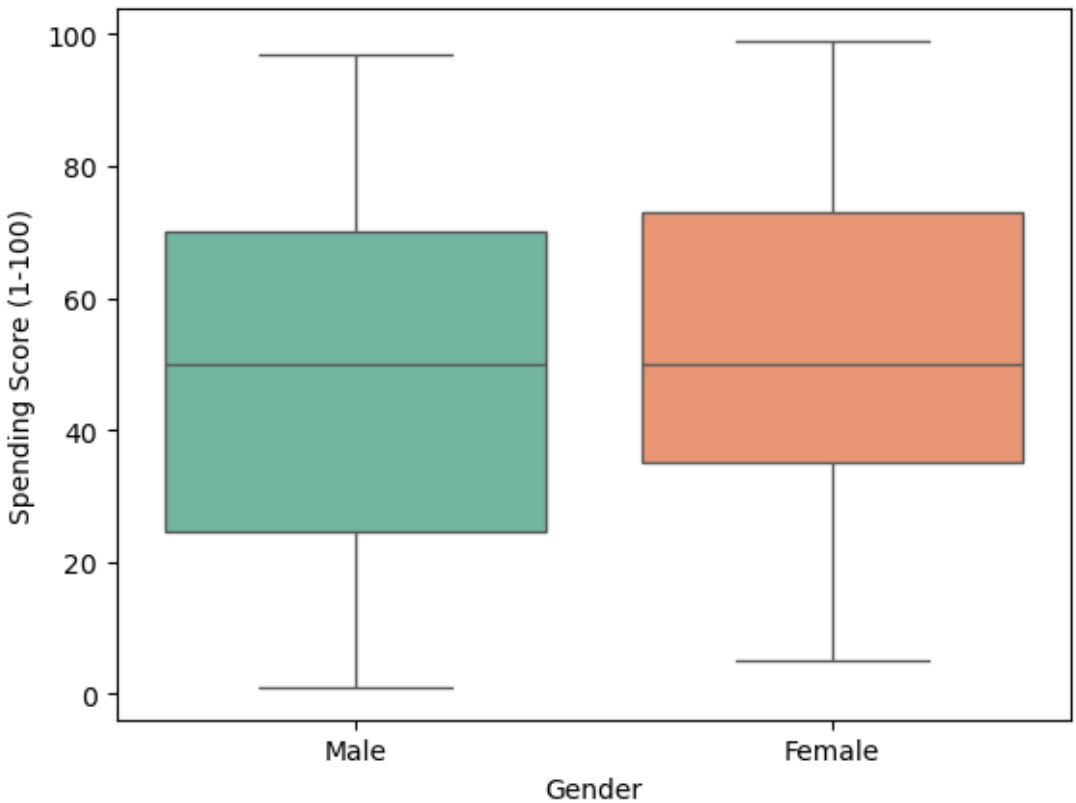


This filter of gender give much more detail to our visualizations and better insights on the mall customers. For example, I can see that between the ages of 20 and 40 there is higher frequency in females. For annual income, between 50k and 80k there is higher frequency in females than males. It also looks like for males there could be an outlier which is suggested by the tail that is formed on the right side. Then looking at spending score, it looks like males have a lower spending score compared to females.

[7]:
```
for i in columns:
    plt.figure()
    sns.boxplot(data=customer_df, x='Gender', y=i, palette='Set2')
```


70
60
50
40
30
20
Age
Male
Female
Gender


140
120
100
80
60
40
20
Annual Income (k$)
Male
Female
Gender



Using boxplots helps me visualize the data in another way. For age I can see that the median is a bit higher compared to females. I can also see that the age in males is more spread out while females is more narrow. The opposite goes for annual income, the median for males is higher but it is most likely due to the outlier and in this boxplot the income is more spread in females than it is in males.

What is interesting in the spending score boxplot is how the median is the same between males and females but the spread is larger in males showing that there are more values in spending scores in males.

[8]:
```
customer_df['Gender'].value_counts(normalize=True)
```

[8]:
```
Gender
Female    0.56
Male      0.44
Name: proportion, dtype: float64
```

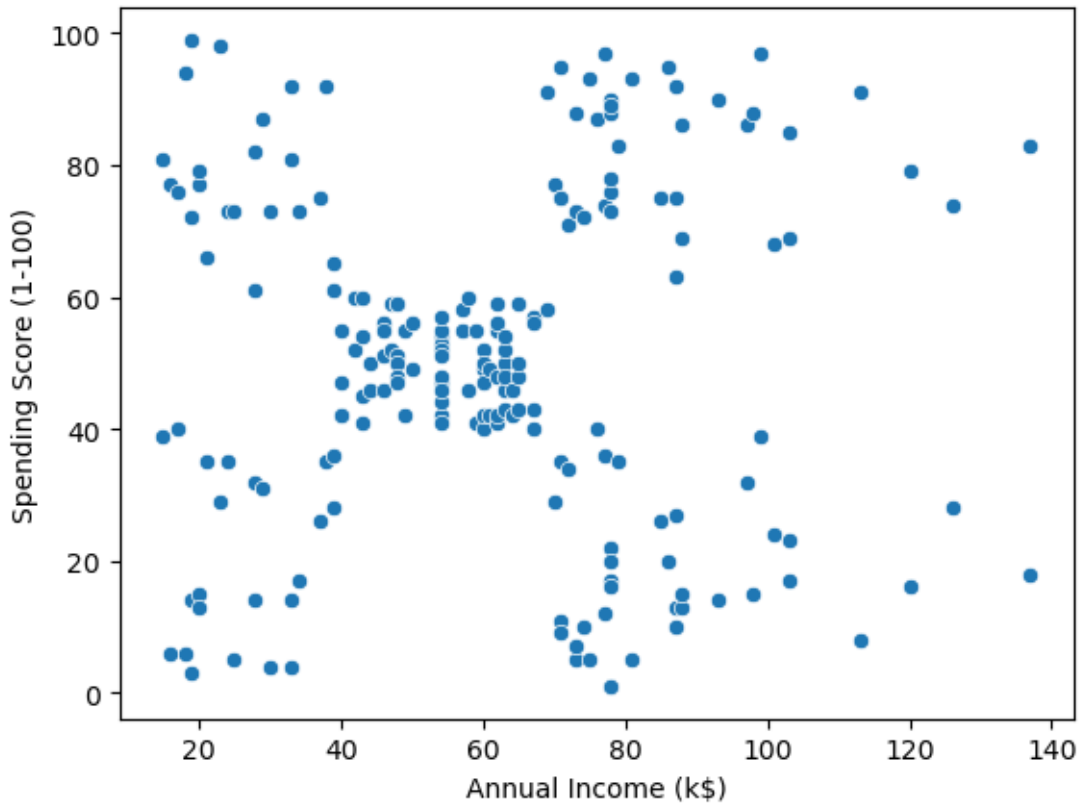
56% of the customers are **female**, and **44%** of them are **male**.

3 Bivariate Analysis

Now I want to look at two variables together.

[9]:
```
sns.scatterplot(data=customer_df, x='Annual Income (k$)', y= 'Spending Score␣
↪(1-100)')
```

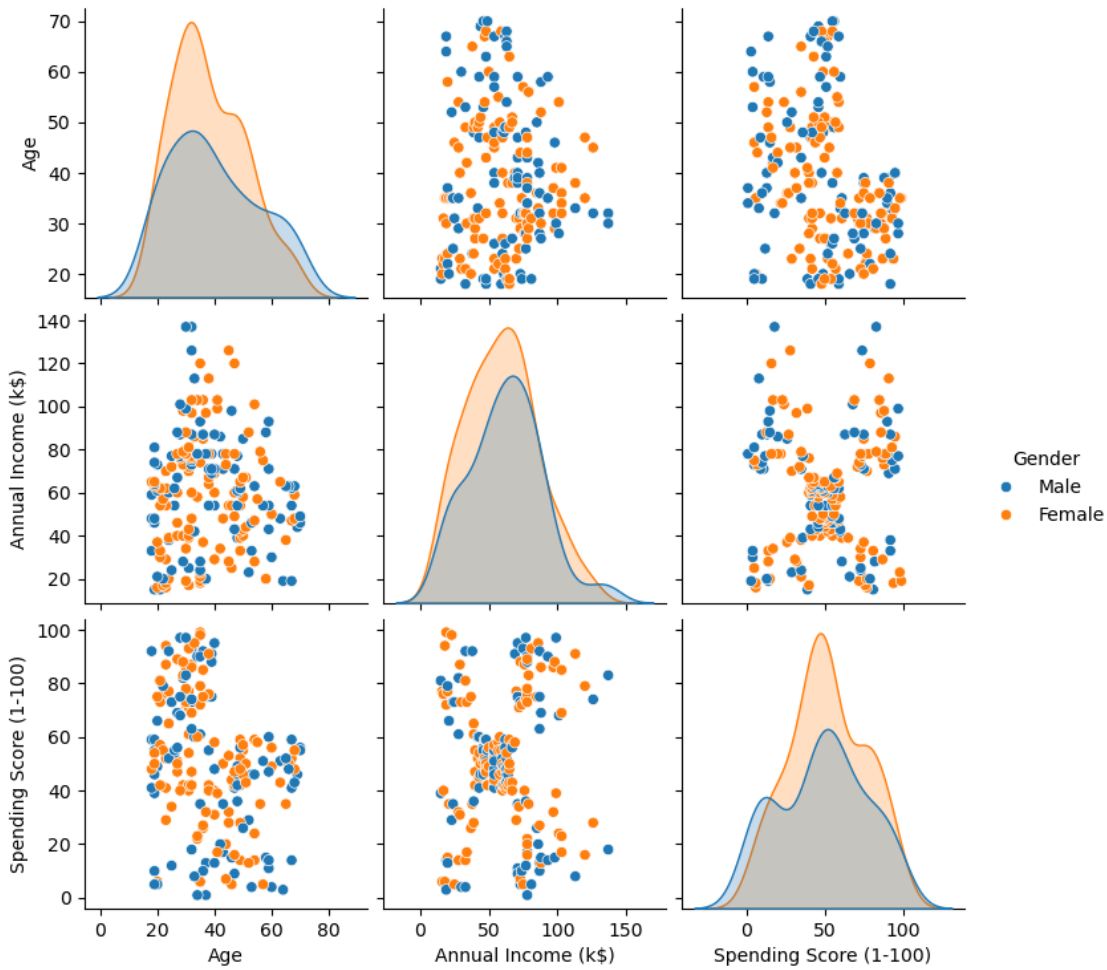
[9]: <Axes: xlabel='Annual Income (k$)', ylabel='Spending Score (1-100)'>



By looking at the scatterplot I can already see some clustering within the data. It looks like there are about 5 clusters in the scatterplot.

[14]:
```
#customer_df = customer_df.drop('CustomerID', axis=1)
sns.pairplot(data=customer_df, hue='Gender')
```

[14]: <seaborn.axisgrid.PairGrid at 0x19541374410>



Using pairplot I made it faster and easier to visualize the relationships between my variables. This helps me identify any clusters quickly, I can see that I will benefit from a bivariate clustering between spending score and annual income.

[12]:
```
customer_df.groupby(['Gender'])[['Age', 'Annual Income (k$)',
        'Spending Score (1-100)']].mean()
```

[12]:

	Age	Annual Income (k$)	Spending Score (1-100)
Gender			
Female	38.098214	59.250000	51.526786
Male	39.806818	62.227273	48.511364

I think it is good to know the mean values of my data based on gender. This gives good information where I can see the average age in male and females is similar. Annual income is slightly higher in males, and spending score is a bit higher in females.

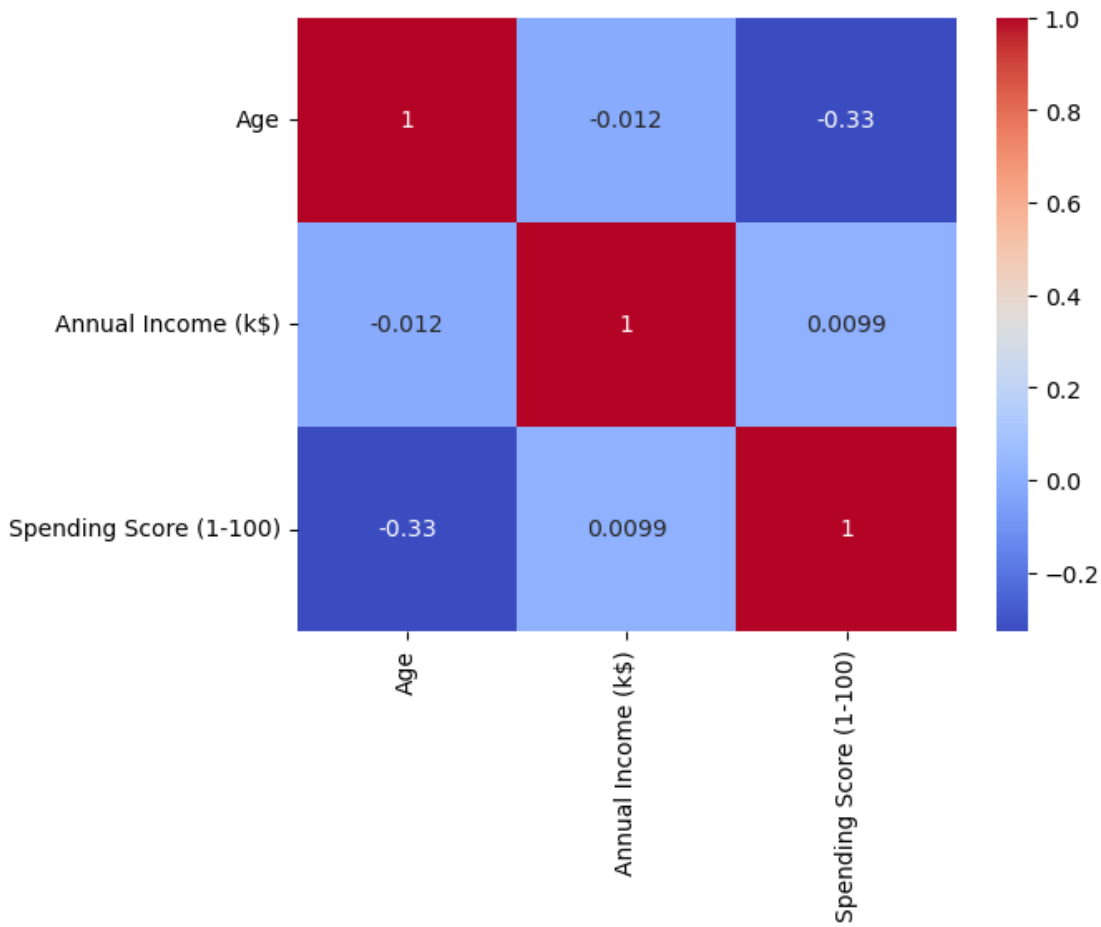
[15]:
```
customer_df.select_dtypes(include='number').corr()
```

[15]:

	Age	Annual Income (k$)	Spending Score (1-100)
Age	1.000000	-0.012398	-0.327227
Annual Income (k$)	-0.012398	1.000000	0.009903
Spending Score (1-100)	-0.327227	0.009903	1.000000

[17]:
```
sns.heatmap(customer_df.select_dtypes(include='number').corr(), annot=True,␣
↪cmap='coolwarm')
```

[17]: <Axes: >



Using a heatmap I can analyze the correlation between two variables. I can see that the relationship between age and annual income has a negative correlation, as well as the relationship between age and spending score. The only positive correlation I see is in the relationship between annual income and spending score. So, **as annual income increases so does spending score. As age increases both annual income and spending score decrease.**

4 Clustering: Unitvariate, Bivariate, Multivariate

I want to find the ideal number of clusters.

[45]:
```
# starting off KMeans had no parameter, I had to identify the number of␣
↪clusters was ideal using the elbow method shown below.
clustering1 = KMeans(n_clusters=3)
```

[46]:
```
clustering1.fit(customer_df[['Annual Income (k$)']])
```

[46]:
```
KMeans(n_clusters=3)
```

[47]:
```
clustering1.labels_
```

[47]:
```
array([2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1], dtype=int32)
```

[48]:
```
customer_df['Income Cluster'] = clustering1.labels_
customer_df.head()
```

[48]:

	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster
0	Male	19	15	39	2
1	Male	21	15	81	2
2	Female	20	16	6	2
3	Female	23	16	77	2
4	Female	31	17	40	2

[49]:
```
customer_df['Income Cluster'].value_counts()
```

[49]:
```
Income Cluster
0    92
2    72
1    36
Name: count, dtype: int64
```

[50]:
```
clustering1.inertia_
```

[50]:
```
23528.152173913048
```

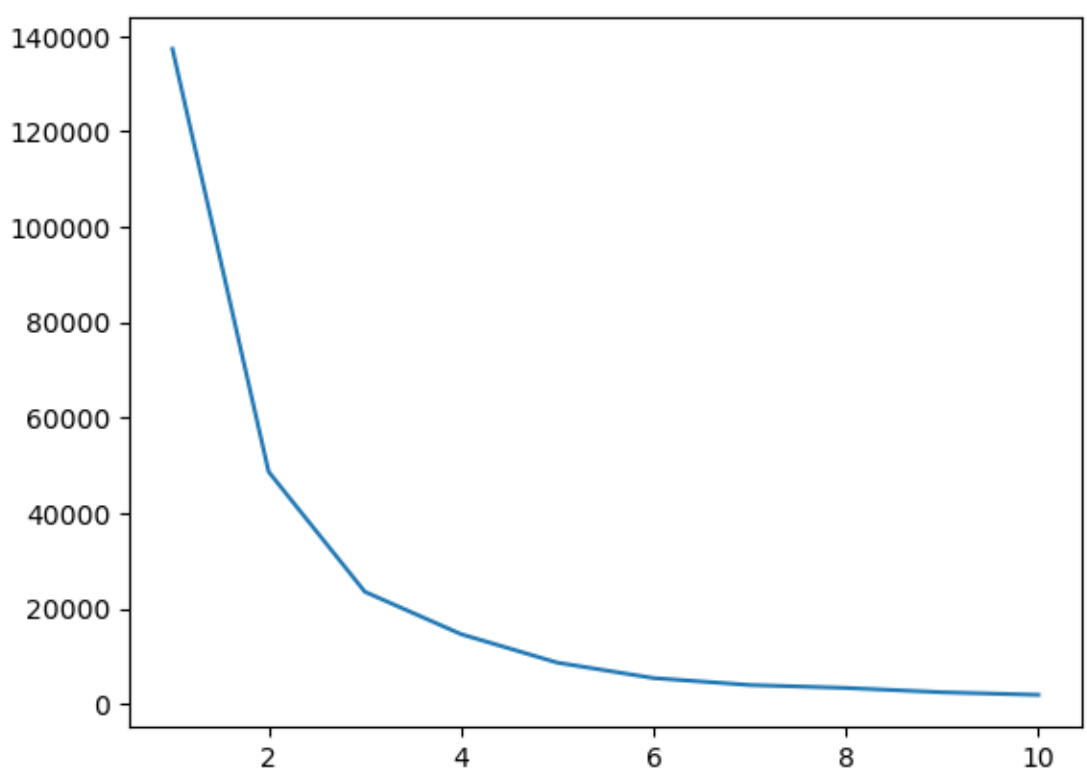
[51]:
```
# using these values I will create a plot for my elbow method to determine how␣
↪many clusters are ideal
inertia_scores = []
for i in range(1,11):
    kmeans = KMeans(n_clusters=i)
    kmeans.fit(customer_df[['Annual Income (k$)']])
    inertia_scores.append(kmeans.inertia_)
```

[52]:
```
inertia_scores
```

[52]:
```
[137277.28000000003,
 48660.88888888888,
 23517.33093093093,
 14631.886877828052,
 8667.679614837509,
 5430.245925925926,
 3984.6157796451917,
 3365.1402470078942,
 2459.137445887446,
 1914.1893162393162]
```

[53]:
```
plt.plot(range(1,11), inertia_scores)
```

[53]:
```
[<matplotlib.lines.Line2D at 0x19541c865d0>]
```



I can see that the elbow starts at 3, so I will go back to my clustering1 variable and change the parameter.

[54]:
```
customer_df.columns
```

[54]:
```
Index(['Gender', 'Age', 'Annual Income (k$)', 'Spending Score (1-100)',
       'Income Cluster'],
      dtype='object')
```

[55]:
```
customer_df.groupby('Income Cluster')[['Age', 'Annual Income (k$)',
       'Spending Score (1-100)']].mean()
```

[55]:

Income Cluster	Age	Annual Income (k$)	Spending Score (1-100)
0	39.184783	66.717391	50.054348
1	37.833333	99.888889	50.638889
2	38.930556	33.027778	50.166667

Seeing these averages helps me understand the data more. I can see that this supports my earlier analysis on how as age increases annual income decreases. I can see that the lowest age has the highest income.

5 Bivariate Clustering

Now I am doing a bivariate clustering. As I mentioned earlier based on the data, a bivariate clustering would benefit me the most. I also know from my earlier analysis that the ideal number of clusters is 5 but I am still going to double check using the elbow test again.

[103]:
```
clustering2 = KMeans(n_clusters=5)
clustering2.fit(customer_df[['Annual Income (k$)', 'Spending Score (1-100)']])
customer_df['Spending and Income Cluster'] = clustering2.labels_
customer_df.head()
```

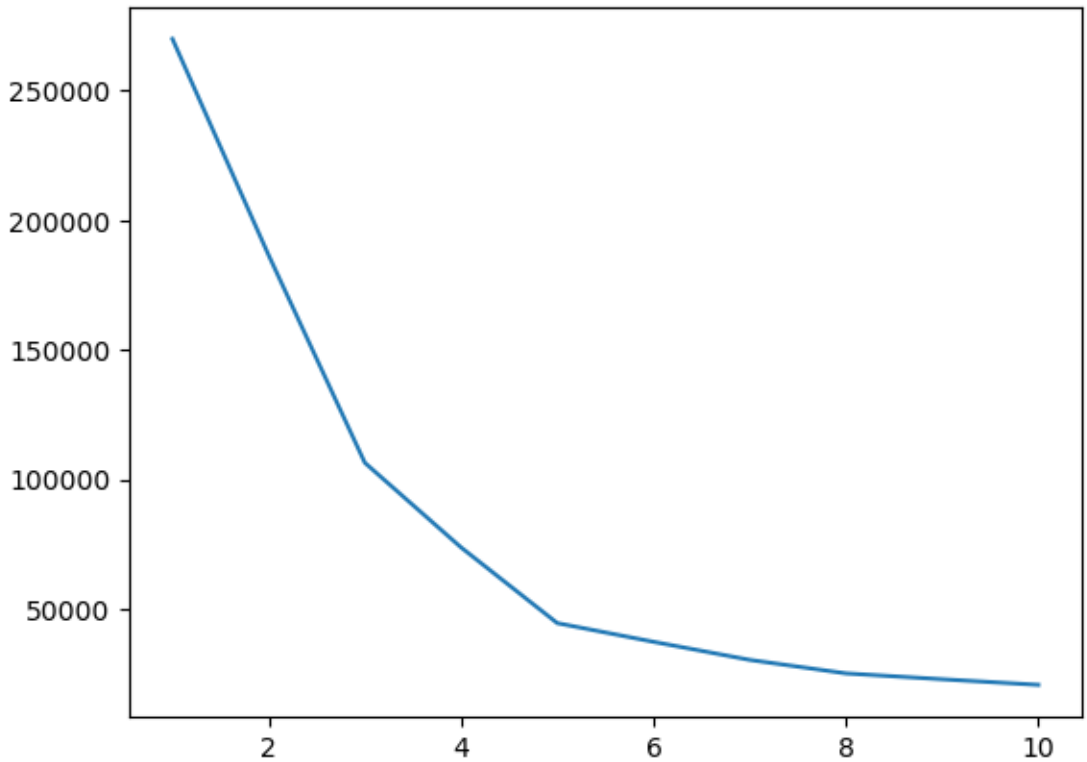
[103]:

	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster	\
0	Male	19	15	39	2	
1	Male	21	15	81	2	
2	Female	20	16	6	2	
3	Female	23	16	77	2	
4	Female	31	17	40	2	

	Spending and Income Cluster
0	4
1	0
2	4
3	0
4	4

[104]:
```
inertia_scores2 = []
for i in range(1,11):
    kmeans2 = KMeans(n_clusters=i)
    kmeans2.fit(customer_df[['Annual Income (k$)', 'Spending Score (1-100)']])
    inertia_scores2.append(kmeans2.inertia_)
plt.plot(range(1,11), inertia_scores2)
```

[104]: [<matplotlib.lines.Line2D at 0x195437534d0>]



Looking at the graph I can see my elbow starts at 5, so I can go back to my clustering2 variable and pass 5 as a parameter.

[105]:
```
centers_df = pd.DataFrame(clustering2.cluster_centers_)
centers_df.columns = ['x', 'y']
centers_df
```

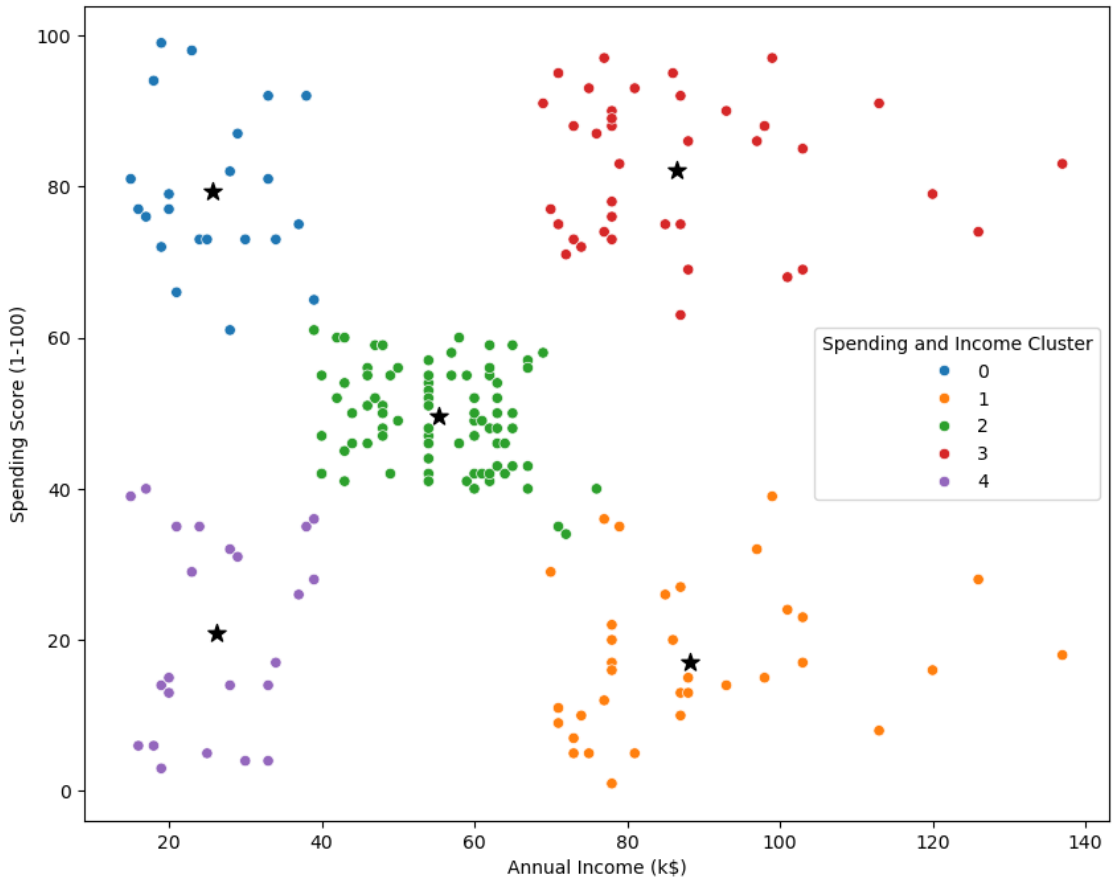
[105]:

	x	y
0	25.727273	79.363636
1	88.200000	17.114286
2	55.296296	49.518519
3	86.538462	82.128205
4	26.304348	20.913043

I want to be able to add these centers to my scatterplot to label the centers of my clusters.

[106]:
```
plt.figure(figsize=(10,8))
plt.scatter(x=centers_df['x'], y=centers_df['y'], s=100, c='black', marker='*')
sns.scatterplot(data=customer_df, x='Annual Income (k$)', y= 'Spending Score␣
↪(1-100)', hue='Spending and Income Cluster',palette='tab10')
```

[106]:
```
<Axes: xlabel='Annual Income (k$)', ylabel='Spending Score (1-100)'>
```



My target cluster will be cluster 3. The one with high spending score and high annual income. This cluster will bring in the most money. We can look further into this cluster and see what gender dominates as well as what the average age is.

[107]:
```
pd.crosstab(customer_df['Spending and Income Cluster'], customer_df['Gender'],␣
↪normalize='index')
```

[107]:

Gender	Female	Male
Spending and Income Cluster		
0	0.590909	0.409091
1	0.457143	0.542857

```
2                           0.592593  0.407407
3                           0.538462  0.461538
4                           0.608696  0.391304
```

This gives me more detail on the clusters and shows me what gender dominates in each. I can see that in my target cluster females dominate over males. Overall in 4 out of the 5 clusters females dominate. Males only dominate in 1.

[108]:
```
customer_df.groupby('Spending and Income Cluster')[['Age', 'Annual Income (k$)',
       'Spending Score (1-100)']].mean()
```

[108]:
```
                                   Age  Annual Income (k$)  \
Spending and Income Cluster
0                           25.272727           25.727273
1                           41.114286           88.200000
2                           42.716049           55.296296
3                           32.692308           86.538462
4                           45.217391           26.304348

                            Spending Score (1-100)
Spending and Income Cluster
0                                        79.363636
1                                        17.114286
2                                        49.518519
3                                        82.128205
4                                        20.913043
```

Here I can see that the average age in my target cluster is 33, the average annual income is 87k, and the average spending score is 82.

6 Multivariate Clustering

[111]:
```
from sklearn.preprocessing import StandardScaler
```

[112]:
```
scale = StandardScaler()
```

[113]:
```
customer_df.head()
```

[113]:
```
   Gender  Age  Annual Income (k$)  Spending Score (1-100)  Income Cluster  \
0    Male   19                  15                      39               2
1    Male   21                  15                      81               2
2  Female   20                  16                       6               2
3  Female   23                  16                      77               2
4  Female   31                  17                      40               2

   Spending and Income Cluster
0                            4
1                            0
```

```
2                            4
3                            0
4                            4
```

[117]:
```
new_df = pd.get_dummies(customer_df, drop_first=True, dtype=int)
new_df.head()
```

[117]:
```
   Age  Annual Income (k$)  Spending Score (1-100)  Income Cluster  \
0   19                  15                      39               2
1   21                  15                      81               2
2   20                  16                       6               2
3   23                  16                      77               2
4   31                  17                      40               2

   Spending and Income Cluster  Gender_Male
0                            4            1
1                            0            1
2                            4            0
3                            0            0
4                            4            0
```

[118]:
```
new_df = new_df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)',␣
 ↪'Gender_Male']]
new_df.head()
```

[118]:
```
   Age  Annual Income (k$)  Spending Score (1-100)  Gender_Male
0   19                  15                      39            1
1   21                  15                      81            1
2   20                  16                       6            0
3   23                  16                      77            0
4   31                  17                      40            0
```

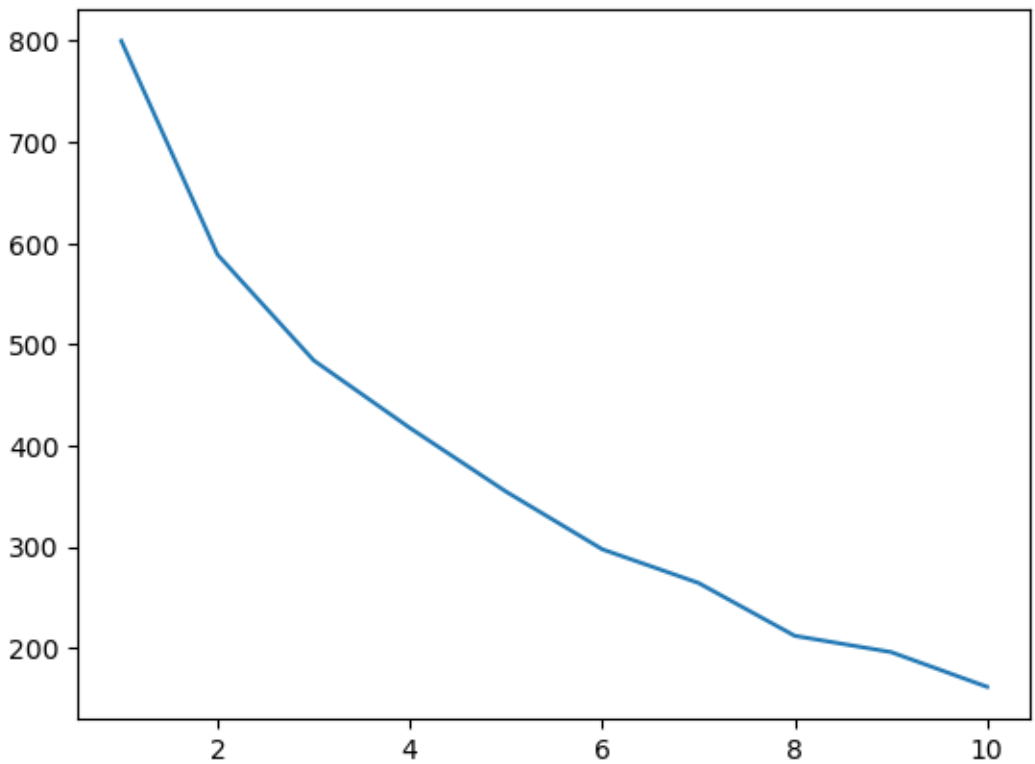
[119]:
```
new_df = pd.DataFrame(scale.fit_transform(new_df))
new_df.head()
```

[119]:
```
          0         1         2         3
0 -1.424569 -1.738999 -0.434801  1.128152
1 -1.281035 -1.738999  1.195704  1.128152
2 -1.352802 -1.700830 -1.715913 -0.886405
3 -1.137502 -1.700830  1.040418 -0.886405
4 -0.563369 -1.662660 -0.395980 -0.886405
```

[121]:
```
inertia_scores3 = []
for i in range(1,11):
    kmeans3 = KMeans(n_clusters=i)
    kmeans3.fit(new_df)
    inertia_scores3.append(kmeans3.inertia_)
```

```
plt.plot(range(1,11), inertia_scores3)
```

[121]: [<matplotlib.lines.Line2D at 0x1953af7cf50>]



I can create a scatterplot but I won't because I think that the bivariate scatterplot gives me great information. Looking back at my goal and objective, using exploratory analysis, univariate and bivariate analysis, as well as clustering I completed the tasks I needed to be done. I identified the target customers and most important shopping groups. I also determined the ideal number of groups which for a univariate clustering and a bivariate clustering.

[ ]:

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